

Which process leads to the formation of benzohydroxamic acid shown in the reaction above?

- ☐ A. Reduction of the amine nitrogen [10%]
☐ B. Carboxylation of the amine nitrogen [13%]
☒ C. Acylation of the hydroxylamine nitrogen [42%]
☐ D. Oxidation of the hydroxylamine nitrogen [33%]

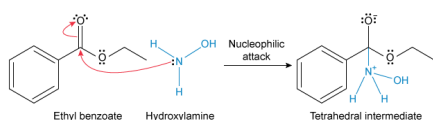
Correct

41% answered correctly

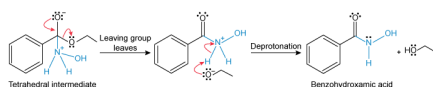
Explanation:

Carboxylic acid derivatives can undergo nucleophilic acyl substitution under acidic or basic conditions. A **nucleophile**, such as an amine, attacks the electrophilic carbonyl carbon of the carboxylic acid derivative, creating a **tetrahedral intermediate**, and the leaving group then leaves.

For example, under basic conditions, **esters** such as ethyl benzoate can undergo nucleophilic attack by an **amine** or, as in this question, a **hydroxylamine**, creating a tetrahedral intermediate with C–N bond formation.



Then the –OR (alkoxy group) on the ester leaves and the carbonyl group is reformed. In this case, the –OR group is –OCH₂CH₃. Finally, the amine is deprotonated by the alkoxy group, yielding a **hydroxamic acid**, benzohydroxamic acid in this case.



The given reaction results in the **addition of an acyl group** –RC=O to the N atom of hydroxylamine. Therefore, the N atom of hydroxylamine undergoes an **acylation reaction**.

(Choice A) The nitrogen-containing compound is hydroxylamine rather than just an amine. Reduction of an atom requires a decrease in the number of bonds to a more **electronegative** atom and an increase in the number of bonds to a less electronegative atom. Because the hydroxylamine N loses a bond to H (less electronegative) and gains a bond to C (also less electronegative), the N atom in hydroxylamine does not lose bonds to a more electronegative atom and therefore is *not* reduced.

(Choice B) Carboxylation would result in the addition of a *carboxyl* group RCOOH rather than an acyl group.

(Choice D) Oxidation of an atom requires an increase in the number of bonds to a more electronegative atom and a decrease in the number of bonds to a less electronegative atom. While the hydroxylamine N loses a bond to H, a less electronegative atom, it gains a bond to C, an atom also less electronegative than N. Because N does *not* gain a bond to an atom *more* electronegative than itself, hydroxylamine was *not* oxidized.

Educational objective:

Hydroxylamine (NH₂OH, an oxygenated version of ammonia) can react with an ester under basic conditions to form a hydroxamic acid. This reaction results in the addition of an acyl group –RC=O to the hydroxylamine and is known as an acylation.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 403427
System:
Functional Groups and Their Reactions